

Muhammad Farhan Shakeel

Electrical Engineering Student | Digital Systems • Computer Architecture • RTL • ASIC/VLSI

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Location Lahore, Pakistan

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Focus Hardware / Semiconductor Engineering

PROFESSIONAL SUMMARY

Electrical Engineering student at the University of Engineering and Technology, Lahore, focused on **digital systems, computer architecture, RTL design, FPGA development and ASIC/VLSI**. Hands-on experience designing and verifying digital hardware with **Verilog/SystemVerilog**, building processor and peripheral designs, and working through RTL-to-GDS physical-design flows. Experienced with **RISC-V, UART, FIFO, memory controllers, FPGA prototyping, PCB design and embedded systems**, supported by programming skills in **C/C++ and Python**. Interested in developing efficient computing hardware and advancing toward professional chip design and computer architecture roles.

TECHNICAL SKILLS

Digital Design	RTL Design, SystemVerilog, Verilog, FSMs, UART, FIFO, Memory Controllers
Computer Architecture	RISC-V / RV32I, Processor Design, Datapaths, Control Logic
ASIC / VLSI	Yosys, OpenROAD, LibreLane, OpenSTA, Magic, RTL-to-GDS Flow, Static Timing Analysis
FPGA	Xilinx Vivado, Artix-7, FPGA Prototyping, Icarus Verilog, GTKWave
PCB & Embedded	KiCad, Altium, ESP32, STM32, Microprocessors, Digital Hardware Integration
Programming & OS	C, C++, Python, Linux, Git

EXPERIENCE

HiCoar — Computer Architecture	2026 – Present
- Designing and developing a RISC-V processor using SystemVerilog, with emphasis on RTL structure and computer-architecture concepts. - Working on RTL design, verification and digital-system integration for processor and hardware modules. - Applying an ASIC physical-design flow including synthesis, floorplanning, placement, routing, static timing analysis and signoff checks. - Using open-source EDA tools including Yosys, LibreLane/OpenROAD, OpenSTA and Magic for hardware implementation and analysis.	

SELECTED PROJECTS

- RISC-V Single-Cycle Processor** — Designed an RV32I single-cycle processor in SystemVerilog with datapath, control, register-file and ALU components; verified through simulation.
- UART Memory Controller** — Implemented UART TX/RX communication with a memory-control FSM for read/write operations and FPGA-based testing.
- ASIC RTL-to-GDS Flow** — Implemented digital designs through synthesis, floorplanning, placement, routing, STA and physical signoff using the Sky130 open-source ecosystem.
- FPGA Digital Systems** — Developed and tested practical FPGA modules including FIFO, UART, PWM/audio control, seven-segment display and sorting/processor logic.

EDUCATION

BS Electrical Engineering	2024 – Present
University of Engineering and Technology (UET), Lahore	
F.Sc. Pre-Engineering	2024
Punjab Group of Colleges, Chishtian — 1023/1200	
Matriculation	2022
Misali Public Science High School, Dahranwala — 1075/1100	

LANGUAGES & INTERESTS

Languages: Urdu • English • German **Interests:** Semiconductor Technology • Chip Design • Computer Architecture • Digital Systems • Problem Solving